

Program 1

Step 1: Create [App.py](#) and requirements.txt

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog1$ cat requirements.txt
flask
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog1$ cat app.py
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello,Welcome to RVCE"

if __name__ == '__main__':
    app.run(host='localhost', port=5000)
```

Step 2: Create Docker file

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog1$ cat Dockerfile
#This is a Dockerfile used to create Docker Images
#Use slim python base image

FROM python:3.9-slim

#Set Working Directory
WORKDIR /app

#Upgrade pip and install system dependencies
RUN apt-get update && apt-get install -y build-essential && \
    pip install --upgrade pip setuptools

#Copy the Requirements.txt file and install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

#Copy the Application code
COPY . .

#set Flask environment Variables
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_PORT=5000

#expose the PORT
EXPOSE 5000

#Command to run the Flask App
CMD ["flask", "run"]
```

Step 3: Create Docker image

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog1$ docker build -t program-1 .
[+] Building 197.4s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 683B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/6] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB
=> => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB
=> => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B
=> => sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b 10.36kB / 10.36kB
=> => sha256:dad5b29e3506c35e0fd22736f4d4ef25d21b219acdd73f7bb41d59996ca8e0d 1.74kB / 1.74kB
=> => sha256:085da638e1b8a449514c3fda83ff50a3bffa4418b050cfacd87e5722071f497 5.40kB / 5.40kB
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08
=> => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508
=> [internal] load build context
=> => transferring context: 93B
=> [2/6] WORKDIR /app
=> [3/6] RUN apt-get update && apt-get install -y build-essential && pip install --upgrade pip setuptools 174.0s
=> [4/6] COPY requirements.txt .
=> [5/6] RUN pip install --no-cache-dir -r requirements.txt 7.9s
=> [6/6] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:b4421e96e82491fde704c1ff65f113a1246dac78b48280cc8eac1bc83a9d6ac3
=> => naming to docker.io/library/program-1
=> 0.0s
```

Step 4: Run Docker image as a container

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog1$ docker run -p 5000:5000 program-1
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [07/Nov/2025 04:17:52] "GET / HTTP/1.1" 200 -
```

Step 5: Output

